```
587 inodes, blocksize: 65536 bytes, created: 2007-06-14
00:34:20
```

Extract

The '-e' option with binwalk will extract the individual files in the firmware as shown below:

```
$ binwalk -e dd-wrt.v24_whr-g125.bin

DECIMAL       HEXADECIMAL     DESCRIPTION
--------------------------------------------------------------
0             0x0             TRX firmware header, little
endian, image size: 3543040 bytes, CRC32: 0x85472C8C, flags:
0x0, version: 1, header size: 28 bytes, loader offset: 0x1C,
linux kernel offset: 0x8E4, rootfs offset: 0x8E7CC
28            0x1C            gzip compressed data, maximum
compression, from Unix, last modified: 1970-01-01 00:00:00
(null date)
2276          0x8E4           LZMA compressed data,
properties: 0x5D, dictionary size: 8388608 bytes, uncompressed
size: 1982464 bytes
```

The list of files can be viewed from the extracted folder:

```
$ ls _dd-wrt.v24_whr-g125.bin.extracted/
1C  8E4  8E4.7z  8E7CC.squashfs  squashfs-root
```

Recursive extraction

You can also perform recursive extraction using the '-M' option with '-e' on the *dd-wrt.v24_whr-g125.bin* firmware as illustrated below:

```
$ binwalk -Me dd-wrt.v24_whr-g125.bin

Scan Time:     2024-08-18 17:24:57
Target File:   /home/guest/dd-wrt.v24_whr-g125.bin
MD5 Checksum:  8a60810685fa5be6221936034b81fd3a
Signatures:    436
...
DECIMAL       HEXADECIMAL     DESCRIPTION
-------------------------------------------------------------

13309         0x33FD          JBOOT STAG header, image id: 2,
timestamp 0x12A200, image size: 134217728 bytes, image JBOOT
checksum: 0x200, header JBOOT checksum: 0x6824
...
280470        0x44796         ESP Image (ESP32): flash mode:
QUIO, flash speed: 40MHz, flash size: 1MB, entry address:
0x90000
...
620573        0x9781D         JBOOT STAG header, image id: 5,
timestamp 0x5008310, image size: 1064960 bytes, image JBOOT
checksum: 0x0, header JBOOT checksum: 0x2000
   ...
1014905       0xF7C79         JBOOT STAG header, image id: 6,
timestamp 0x803E3, image size: 83886080 bytes, image JBOOT
checksum: 0xA000, header JBOOT checksum: 0x10
...
1632592       0x18E950        CRC32 polynomial table, little
endian
1634928       0x18F270        Linux kernel version 2.4.34
1648720       0x192850        Unix path: /usr/lib/libc.so.1
...
```

Extract <type> signature

If you would like to extract a specific signature type file, you need to specify the same with the '-D' option. For example, the following command extracts all .7z compressed files from the input firmware file.

```
$ binwalk -D '7z' dd-wrt.v24_whr-g125.bin

DECIMAL       HEXADECIMAL     DESCRIPTION
-------------------------------------------------------------
-------------------
0             0x0             TRX firmware header, little
endian, image size: 3543040 bytes, CRC32: 0x85472C8C, flags:
0x0, version: 1, header size: 28 bytes, loader offset: 0x1C,
linux kernel offset: 0x8E4, rootfs offset: 0x8E7CC
28            0x1C            gzip compressed data, maximum
compression, from Unix, last modified: 1970-01-01 00:00:00
(null date)
2276          0x8E4           LZMA compressed data,
properties: 0x5D, dictionary size: 8388608 bytes,
uncompressed size: 1982464 bytes
583628        0x8E7CC         Squashfs filesystem, little
endian, DD-WRT signature, version 3.0, size: 2954340 bytes,
587 inodes, blocksize: 65536 bytes, created: 2007-06-14
00:34:20

$ ls _dd-wrt.v24_whr-g125.bin.extracted/
8E4  8E4.7z
```

Hexdump

The '-W' option provides a hexdump of the binary file as shown below:

```
$ binwalk -W dd-wrt.v24_whr-g125.bin

OFFSET      dd-wrt.v24_whr-g125.bin
-------------------------------------------------------------
0x00000000  48 44 52 30 00 10 36 00 8C 2C 47 85 00 00 01 00
|HDR0..6..,G.....|
0x00000010  1C 00 00 00 E4 08 00 00 CC E7 08 00 1F 8B 08 00
```